

SAPP4VU: CLOUD MASK

Implementazione di algoritmi numerico-statistici per la caratterizzazione e rimozione del rumore e per la cloud detection in immagini iperspettrali.



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PRISMA CLOUD MASK - GIT

DESCRIPTION:

This repository contains the Python sources of the Prisma basic processing for cloud classification. Some parts for the preprocessing were adapted from the original code developed by [1] - see the **ACOLITE: generic atmospheric correction module - for PRISMA**

PREPARE ENVIRONMENT

-Example based on linux systems-

1. Create an environment, for instance:

```
$ pip install virtualenv  
$ python -m venv <virtual-environment-name>
```

or if necessary:

```
$ pip3 install virtualenv  
$ python3 -m venv <virtual-environment-name>
```

2. Activate your virtual environment:

```
$ source <virtual-environment-name>/bin/activate
```

3. Install the requirements in the Virtual Environment, you can easily just pip install the libraries. For example:

```
$ pip install numpy
```

or If you use the requirements.txt file:

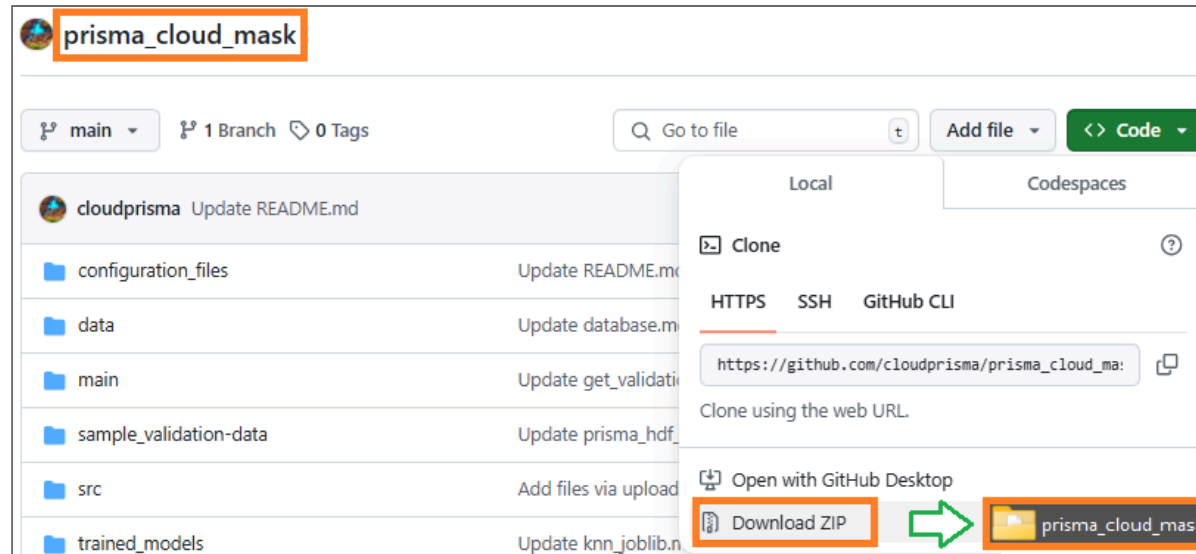
```
$ pip install -r requirements.txt
```

DOWNLOAD PROJECT

-Resources-

4. Download the scripts available here and save them into the same directory or try via git clone source:

```
$ git clone https://github.com/cloudprisma/prisma_cloud_mask
```



Alternative you can download the zip. Please make sure to rename the directory as: **prisma_cloud_mask** after unzip the files.

Given its weight, some files are attached as google drive link. Do not forget to download them:

- Database
- sample validation data/gpkg/ folder
- sample validation data/hdf/ folder
- trained models/knn.joblib

RUN THE SCRIPTS

-To get the Cloud Mask-

STEP 1:

- Configure the config_getclassification.json file according to the instructions given in this link.

CONFIG_GETCLASSIFICATION.JSON

This is an example for the configuration file used to classify a new image. It is called by the **get_classification.py** main script. The structure for this file is described below:

```
{  
  "img_folder": "../hdf/",  
  "trained_models_folder": "../trained_models/",  
  "output_mask_folder": "../results/",  
  "model": "xgboost",  
  "stack_tif": true,  
  "cloud_prisma": true,  
  "npy_matrix": false,  
}
```

MANDATORY PARAMETERS:

- **img_folder**: path to the images to classify in hdf5 format.
- **trained_models_folder**: path to the trained model in joblib format
- **output_mask_folder**: path to the output directory
- **model**: model to be used, options are: knn, rf (Random Forest) or xgboost

OPTIONAL PARAMETERS:

- **stack_tif**: boolean parameter. If true, the original bands are going to be stacked to the output mask.
- **cloud_prisma**: boolean parameter. If true, the original prisma cloud mask is going to be stacked to the output mask.
- **np_matrix**: boolean parameter. If true, a numpy array is also generated in the output folder.

STEP 2:

- Locate at the directory called: **main** inside of **prisma_cloud_mask** folder. Once there, execute the next command in a terminal, for example, to run the classification script (get_classification.py), you can run the following line:

```
$ python get_classification.py -i <Path to the  
config_getclassification.json file>
```

SAMPLE RESULTS

- Result based on: PRS_L1_STD_OFFL_20240522095507_20240522095511_0001 - Xgboost

